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"INFLUENCE OF REPLACEMENT OF RECYCLED AGGREGATE ON STRENGTHS OF CONCRETE"

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By

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NOMENCLATURE

GDP	- Gross Domestic Product
GCF	- Gross Capital Formation
PPP	- Public Private Partnership
CPHEEO	- Central Public Health & Environmental Engineering Organization
C&D	- Construction and Demolition
RCA	- Recycled Concrete Aggregate
CTM	- Compression Testing Machine
UTM	- Universal Testing Machine
FA	- Fine Aggregate
CA	- Coarse Aggregate
IS	- Indian Standard

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ABSTRACT

In our present society the major challenges is protection of environment. Cities are suffering because of their own Construction and Demolition Waste (CDW) with serious environmental consequences. Reducing, recycling and re-using appear to be the best option.

According to Central Public Health & Environmental Engineering Organization (CPHEEO), India produces 10-12 million tons of CDW per annum. Amongst which Soil, Sand & Gravels is 36%, Brick & Masonry is 31%, 23% of Concrete, 5% Metals, 2% each o0f Bitumen and Wood and 1% for others. India is even facing a shortage of about 55,000 million cubic meters of aggregates for just the real estate industry. Also in road construction sector about 750 million cubic meters of aggregate is required to achieve the target. If there are no favourable uses, the materials become wastes and there will be a large environmental and financial pressure on the authorities to treat and dispose them.

Previous researches on this topics has stated the techniques for producing concrete using recycled aggregate obtained from CDW. Also the effects were studied on the compressive strength of concrete conducting various laboratory trials, replacing the natural aggregate by the recycled ones. Replacement of aggregates was done in percentage. Results showed that lower level replacement had little effects on the compressive strength, whereas higher level of replacement reduces the compressive strength. Concrete deterioration may be due to many reasons: Corrosion of enclosed metals, Chemical attack, Alkali-Aggregate reactivity, Fire, Overloaded, etc.

So in this project Concrete is going to be tested made from recycled aggregates replacing natural aggregates. Such that the recycled aggregates are sub-grades with respect to the service life of the parent concrete, i.e. concrete containing recycled aggregates with 1 - 20years, 21 - 40years and above 40years of service life is going to be tested. With every 10% additive percentage variations.

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CHAPTER 1

INTRODUCTION

1.1 Construction in India

In India the construction creates investment opportunities across various related sectors and though is an important indicator of the development. The industry is fragmented, with a handful of major companies involved in the construction activities across all segment; medium companies broaden in ambience activities; and small contractors who work on the subcontractor basis and carry out the work in the field. The sector is labor-intensive and, including indirect jobs, provides employment to more than 35 million people and creates assets worth over Rupees 200 billion.

The period from 1970 to mid 60's witnessed the government playing an active role in the development of these services and most of construction activities during this period were carried out by state owned enterprises and supported by government departments.

In India Construction has accounted for around 40 per cent of the development investment during the past 50 years. Nation's 16 per cent of working population depends on construction for its bread and butter. It contributes more than 5 per cent to the nation's gross domestic product (GDP) and 78 per cent to the gross capital formation (GCF). The share of the Indian construction sector in total GFC came down from 60 per cent in 1970-71 to 34 per cent in 1990-91. Thereafter, it increased to 48 per cent in 1993-94 and stood at 44 per cent in 1999-2000. In 21st century, there has been an increase in the share of the construction sector in GDP and GFC.

The construction industry in India covers 200 firms in the corporal sector. In addition, there are about 120,000 Class A contractor registered with various government construction bodies. There are thousands of small contractors, which race for small jobs or work as subcontractors of major or other contractors. Total sales of construction

industry have reached Rupees 428854 million in 2004-05 from Rupees 214519 million in 2000-01.

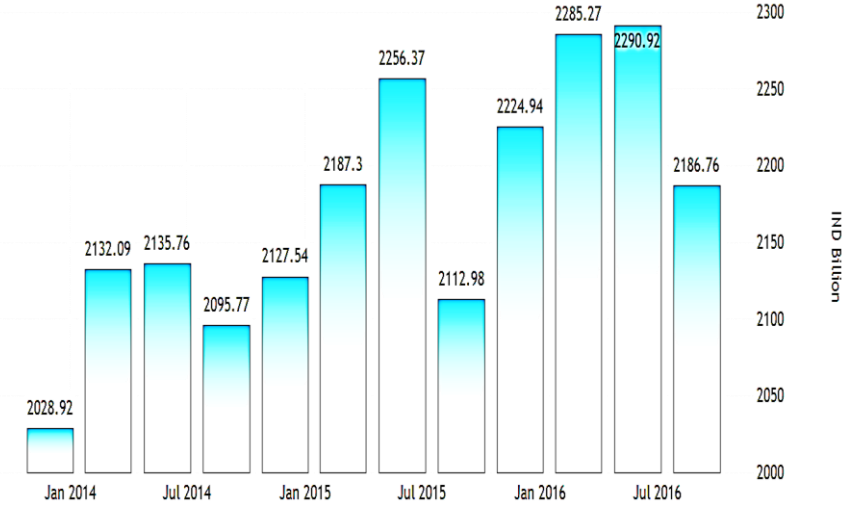


Figure No. 1.1India GDP form Construction
Source: www.tradingeconomics.com | Central Statistic Organization, India

The Indian economy has witnessed considerable progress in the past few decades. Most of the infrastructure development sectors moved forward, but not to the required extent of increasing growth rate up to the tune of 8 to 10 per cent. With the present emphasis on creating physical infrastructure, massive investment is planned in this sector. The planning commission has estimated that investment requirement in infrastructure to the tune of about Rupees 14,500 billion or US$320 billion during the 11th Five Year Plan Period. This is a requirement of an immense magnitude. Budgetary source cannot raise this much resources. Public Private Partnership (PPP) approach is best suited for finding the resources.

1.2 Building Material

Many types of Building materials are used in the building and construction industry to create building and structures. These categories of material and products are used by architects and construction project managers to specify the materials and methods used for building projects. Some building materials like cold rolled steel framing are considered modern methods of construction, over the traditionally slower methods

like block work and timber. Many building materials have a variety of uses; therefore it is always a good idea to consult the manufacturer to check if a product is best suited to your requirements.

Concrete, one of the most widely used construction material, has been subjected to major research and development over the past century. Once thought to provide sound structural and durable performance with little else, concrete has now evolved into a material that is capable of being suitable for environment.

Through the development of high strength concrete during the early sixties for rapid urbanization and high strength requirement, the performance based on concrete mix design was established. High strength concretes are now not only used for their strength capabilities but also for the enhanced durability characteristics.

1.3 Demolition

Since the beginning of the twentieth century, our Nation's infrastructure has grown tremendously. Much of the core infrastructure, including roads, bridges, water systems, and sewers, was put in place during the first half of this century. The Interstate Highway System was constructed during the 1950's, 1960's, and 1970's. Much of this infrastructure has now deteriorated to a point that extensive repair or replacement is required. In areas of rapid population growth, new infrastructure is necessary to meet growing needs.

Any construction activity requires several materials such as concrete, steel, brick, stone, glass, clay, mud, wood, and so on. However, the cement concrete remains the main construction material used in construction industries. For its suitability and adaptability with respect to the changing environment, the concrete must be such that it can conserve resources, protect the environment, economize and lead to proper utilization of energy. To achieve this, major emphasis must be laid on the use of wastes and byproducts in cement and concrete used for new constructions.

The enormous quantities of demolished concrete are available at various construction sites, which are now posing a serious problem of disposal in urban areas. This can easily be recycled as aggregate and used in concrete. Research & Development activities have been taken up all over the world for proving its feasibility, economic viability and cost effectiveness.

The main reasons for increase of volume of demolition concrete / masonry waste are as follows:-

1. Many old buildings, concrete pavements, bridges and other structures have overcome their age and limit of use due to structural deterioration beyond repairs and need to be demolished;
2. The structures, even adequate to use are under demolition because they are not serving the needs in present scenario;
3. New construction for better economic growth;
4. Structures are turned into debris resulting from natural disasters like earthquake, cyclone and floods etc.
5. Creation of building waste resulting from manmade disaster/war.



Figure No. 1.2(a)



Figure No. 1.2(b)

Figure No. 1.2(a) and 1.2(b) Illegal constructions demolitions in PCMC

The use of recycled aggregates from construction and demolition wastes is showing prospective application in construction as alternative to primary (natural) aggregates. It conserves natural resources and reduces the space required for the landfill disposal.

One of the major challenges of our present society is the protection of environment. Some of the important elements in this respect are the reduction of the consumption of energy and natural raw materials and consumption of waste materials. These topics are getting considerable attention under sustainable development nowadays.

1.4 Present Scenario

India produces 10-12 million tons of Construction, Demolition, and Excavation waste material annually. The traditional practice in India is to dispose of this waste in landfills or illegally dump in rivers and water bodies.

As per the Central Public Health & Environmental Engineering Organization (CPHEEO), the Indian Real Estate Industry alone is facing a shortage of aggregates to the extent of 55,000 million cum. In addition, 750 million cum of aggregates would be

required to achieve the targets of road construction sector, which will lead to tremendous pressure on natural resources.

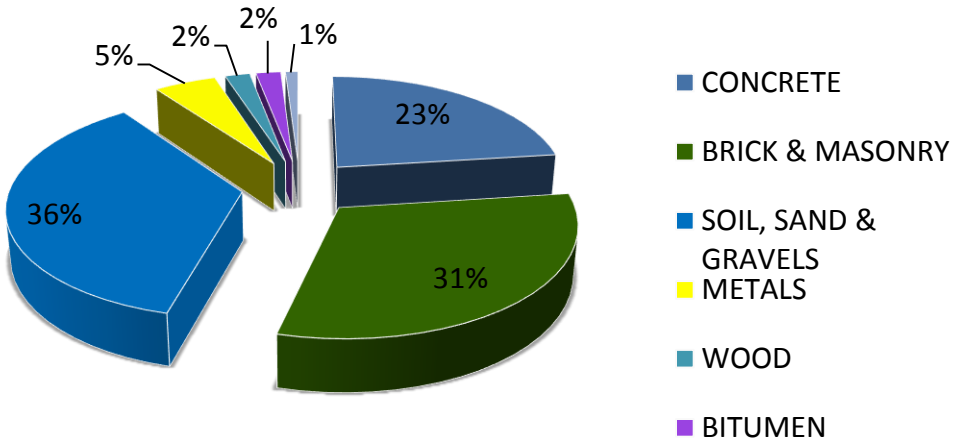


Figure No. 1.3 Composition of Construction & Demolition Waste
Source: Mr. Chiranjib Banerjee, CDE Asia Limited, C&D Waste Processing in India – Delhi Shows the Way, NBM&CW December 2015.

Delhi alone produces more than 4000 tons of C&D Waste and due to improper disposal; this waste is choking storm water drains and polluting the Yamuna river bed.

Leaving aside the current economic crisis, the concrete production is very important. This fact means that the aggregates, the main component of concrete, are much demanded.

Because of the fact that, in some countries, the natural aggregates are a scarce resource, and also because the surface extractions destroy the landscape and upset the biological balance, it is necessary to find a new source of aggregates to guarantee the concrete production. The new source has accomplished two main goals: on the one hand, to maintain the price of concrete, which until now was the cheapest construction material; and on the other, to preserve the environment, without creating piles of waste over the world.

Because it is necessary a new source of aggregates I hope that this research will be a small contribution in the advance of the utilization of the recycled aggregates as construction material in the future.



Figure No. 1.4 Lack of Govt. approved sites for dumping construction debris
Source: H. S. Manjunath, THE HINDU, Mangaluru, October 9 2015.

1.5 Recycled Aggregates

The two most common forms of recycled aggregate used in concrete are glass and crushed concrete from previous applications. While other materials such as fiberglass and granulated coal ash have been experimented with, these two products seem to be the most effective recycled material for use in concrete. Some of the criteria that must be met are the recycled aggregate's ability to provide strength and durability to the finished concrete product as well as offer comparative workability to those mixtures that do not contain any recycled materials. Crushed glass and previously used concrete are also used for their superior hardness properties, in addition to their resistance to absorbing chemicals and compounds commonly found in other materials such as clay and limestone.

The use of crushed glass is often used in concrete applications in which the finished product will not be required to withstand large amounts of weight and pressure.

Glass materials most frequently used for concrete are those that are unable to be recycled in more traditional ways, including light bulbs and glass bottles marked as non-recyclable. In order to be prepared for use in concrete, the glass is finely ground and added to the concrete mixture in place of sand. This concrete is then suited for applications like walkways and biking paths, in addition to any non-structural project needs.

Recycling of concrete is a relatively simple process. It involves breaking, removing, and crushing existing concrete into a material with a specified size and quality. The quality of concrete with RCA is very dependent on the quality of the recycled material used. Reinforcing steel and other embedded items, if any, must be removed, and care must be taken to prevent contamination by other materials that can be troublesome, such as asphalt, soil and clay balls, chlorides, glass, gypsum board, sealants, paper, plaster, wood, and roofing materials.



Figure No. 1.5 Recycled Aggregate

1.6 Recycled Aggregate Characteristics

The crushing characteristics of hardened concrete are similar to those of natural rock and are not significantly affected by the grade or quality of the original concrete. Recycled concrete aggregates produced from all but the poorest quality original concrete can be expected to pass the same tests required of conventional aggregates.

Recycled concrete aggregates contain not only the original aggregates, but also hydrated cement paste. This paste reduces the specific gravity and increases the porosity compared to similar virgin aggregates. Higher porosity of RCA leads to a higher absorption.

1.7 Sustainability

Recycling concrete provides sustainability in several different ways. The simple act of recycling the concrete reduces the amount of material that must be landfilled. The concrete itself becomes aggregate and any embedded metals can be removed and recycled as well. As space for landfills becomes premium, this not only helps reduce the need for landfills, but also reduces the economic impact of the project. Moreover, using recycled concrete aggregates reduces the need for virgin aggregates. This in turn reduces the environmental impact of the aggregate extraction process. By removing both the waste disposal and new material production needs, transportation requirements for the project are significantly reduced.

In addition to the resource management aspect, recycled concrete aggregates absorb a large amount of carbon dioxide from the surrounding environment. The natural process of carbonation occurs in all concrete from the surface inward. In the process of crushing concrete to create recycled concrete aggregates, areas of the concrete that have not carbonated are exposed to atmospheric carbon dioxide.

1.8 Public Perception and Costs

Still today, people have a point of view that, recycled aggregate derived from construction and demolition waste are of inferior quality.

Besides technical considerations and environmental benefits, there are other issues affecting the acceptance, i.e. liability and confidence.

The cost of sending construction and demolition waste to landfill can be greater than the cost of sorting and selling it from construction site to a recycler. It is more effective when the demolished materials are used in a new construction on the same site.

The cost effectiveness for the use of recycled concrete depends on several factors:

1. Proximity and quantity of available natural aggregate
2. Reliability of supply (availability of materials and capacity of recycling facility)
3. Public perceptions regarding the quality of recycled products
4. Standards and regulations requiring different treatment for recycled aggregate compared to primary material

1.9 Objective and Scope of Present Research

The present research aimed to investigate the behavior of concrete with various percentage of recycled aggregate and to investigate the following properties. In view of above discussion the present proposal has the following broad objective.

1. Compare the basic properties of concrete made with natural aggregate and that of recycled aggregate.
2. To find out the safe percentage replacement of natural aggregate by recycled aggregate.
3. To find out the ways of cost saving such as transportation, excavation etc.
4. To reduce the impact of waste materials on environment.
5. To propose the best suitable building configuration on the existing condition**.**
    a. The potential to save natural resources (stone, river sand, soil etc.) and energy, exits in these wastes.
    b. It's occupying significant space at landfill sites.
    c. Its presence spoils processing of bio-degradable as well recyclable waste, Construction & Demolition waste has potential use after processing and grading, Utilization of Construction & Demolition waste is quite common in industrialized countries but in India so far no organized effort has been made.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Recently it has been observed that there is considerable need in improving the handling of construction and demolished waste as it's usually dumped in desolated areas. Construction materials are increasingly judged by their ecological characteristics. Concrete recycling gains importance because it protects natural resources and eliminates the need for disposal by using the readily available concrete as an aggregate source for new concrete or other application.

In this chapter the previous experimental research works carried out regarding Recycled Aggregate are reviewed and certain conclusion are drawn based on which the experimental program for current experimental study is finalized.

2.2 Experimental And Analytical Studies

Margaret Mary O'Mahony[3] (1990) conducted a major part of work with a view to using the recycled aggregate in unbound condition e.g. as road sub-base or fill to structure. In addition, a study was carried out on the properties of concrete made using crushed concrete and the performance of recycled aggregate concrete was compared with that of concrete made using Thames valley gravel.

The 8% increase in free water in the recycled aggregate mixes, was too high to achieve a workability similar to that of the controlled concrete.it is likely that the water allowed in the mixes for water absorption of the aggregate was not completely used because the aggregate was not pre-soaked before mixing. At early ages, the recycled aggregate concretes had lower compressive strengths but by 28 days had achieved the strengths of corresponding controls at more consistent rates.

David R. Wilburn et. al.[5] (1998) studied about the costs for producing recycled aggregates were developed based on data from the Denver, Colo., area. Costs in other regions of the United States may differ due to raw material supply, operational, competition, or demand variations. The methodology used in this study allows for variation in costs and revenues to be evaluated and analyzed.

Both product price and tipping fees vary significantly from region to region, as each region has different market conditions, specifications, and aggregate sources. Because crushed stone and sand and gravel contribute a far greater volume of source material to the aggregates market than recycled material in most regions, the price of available crushed stone or sand and gravel often determines the local market price for recycled aggregates. In a similar manner, local tipping fees are set based upon the volume of waste material, the availability of disposal sites, local demand for aggregates, relative transportation distances and costs, and legislation.

The Government of Hong Kong special administrative region[1] (2007) studied, the properties of different mixes of concrete produced with natural aggregate and produced using recycled aggregate, with 20% and 100% replacement by weight, and with 0% and 35% PFA replacement. Concrete cubes, concrete cylinders and concrete prisms were cast for testing. The compressive strength of concrete at 7 and 28 days of age of concrete was determined. Accelerated bulk diffusion and profile grinding test were adopted to study chloride penetration. Shrinkage test was used to monitor the dimensional change of the specimens. Slump test was measured to indicate the change of workability. Other general properties of concrete were also studied such as young modulus of elasticity, water absorption and shrinkage.

The results showed that there is no noticeable difference of the properties of concrete with 20% recycled aggregate from that of natural aggregate concrete. However, the difference is significant for concrete with 100% recycled aggregate replacement.

Cement Concrete & Aggregates Australia[7] (2008) presented a report clearly indicates an increasing trend and incentive for the greater use of manufactured and

recycled aggregates in construction. There are, however, limitations to the use such materials. This report focuses on known benefits and limitations of arrange of manufactured and recycled aggregates. Successful strategy must be based on both cost and performance.

In all cases, the availability and consistency of supply are prerequisite for the use of manufactured and recycled aggregates in the various applications.

Mirjana Malešev et. al.[8] (2009) verified the type and quantity of coarse aggregate in the following way: -

1. Concrete mix had 100% of natural river coarse aggregate (R0), control mixture,
2. Concrete mix had 50% of natural river coarse aggregate and 50% of recycled coarse aggregate (R50)
3. Concrete mix had 100% of recycled coarse aggregate (R100)

Tested concrete types (R0, R50 and R100) were used for producing RC beams. Three beams with a length of 3.0 m and rectangular cross section of 15/25 cm were prepared for flexural testing. Beams were reinforced with ribbed reinforcement 3RØ12 in the lower zone, 2RØ10 in the upper zone and with stirrups Ø6/20.

Concrete compressive strength mainly depends on the quality of recycled aggregate. If good quality aggregate (obtained by crushing higher strength class concrete as in this case) is used for the production of new concrete, the recycled aggregate has no influence on the compressive strength, regardless of the replacement ratio of natural coarse aggregate with recycled aggregate. The same conclusion is valid for concrete tensile strength (splitting and flexural).

Dr. Chetan D.Modhera et. al.[4] (2011) concluded that concrete strength decreases when recycled concrete was used and the strength reduction could be as low as 40%. However, no decrease in strength was reported for concrete containing up 20% fine or 30% coarse recycled concrete aggregates, but beyond these levels, there was a systematic decrease in strength as the content of recycled aggregates increased. The

strength characteristics of concrete was not affected by the quality of recycled aggregate at high water/cement ratio, it was only affected when the water/cement ratio is low. The higher the water/cement ratio, the less reduction in compressive strength.

There is significant potential for growth of recycled and secondary aggregates as an appropriate and "green" solution to the anticipated increased world – wide construction activity and with it the demand for RSA.

Dr. Sunil S. Pimplikar[2] (2012) performed an experiment on the material collected from sites (i.e. Sri Savitribai Phule Polytechnic, Pune &Mangaol Grampanchyat School, Mangaol, Haveli, Pune) respectively. This collected material is crushed by hammer to separate the aggregates & reduce their sizes in smaller fraction. On these separated aggregates various testes are conducted in laboratory as per Indian Standard code & their results are compared with natural aggregates. Recycled aggregate reduces the impact of waste on environment. By using some percentage in construction sector, cost is saved, due to reduction of transportation & manufacturing process.

Use of recycled aggregate up to 30% does not affect the functional requirements of the structure as per the findings of the test results. Various tests conducted on recycled aggregates and results compared with natural aggregates are satisfactory as per IS 2386. Due to use of recycled aggregate in construction, energy & cost of transportation of natural resources & excavation is significantly saved. This in turn directly reduces the impact of waste material on environment.

Ashraf M. Wagih et. al.[10] (2012) discusses the possibility to replace natural coarse aggregate (NA) with recycled concrete aggregate (RCA) in structural concrete. An investigation into the properties of RCA is made using crushing and grading of concrete rubble collected from different demolition sites and landfill locations around Cairo. A total of 50 concrete mixes forming eight groups were cast. Groups were designed to study the effect of recycled coarse aggregates quality/content, cement dosage, use of super plasticizer and silica fume. Tests were carried out for: compressive strength, splitting strength and elastic modulus.

The results showed that the concrete rubble could be transformed into useful recycled aggregate and used in concrete production with properties suitable for most structural concrete applications in Egypt. A significant reduction in the properties of recycled aggregate concrete (RAC) made of 100% RCA was seen when compared to natural aggregate concrete (NAC), while the properties of RAC made of a blend of 75% NA and 25% RCA showed no significant change in concrete properties.

Kho Pin Verian et. al.[9] (2013) studied that the compressive strengths of all mixtures increased with age. The concrete mixture with 30% RCA had the highest compressive strength at all ages while the mixture with 100% RCA and no fly ash had the lowest compressive strength at all ages beyond 7 days. It should be noted that the higher w/cm (0.47) undoubtedly contributed to these lower strengths. The plain concrete mixture with 50% RCA had strengths similar to those of the control mixtures without RCA.

For concrete with 30% RCA, the use of fly ash did not improve the compressive strength. However, for concretes with 50% and 100% RCA, the use of fly ash in the mixture did improve the compressive strength after 7 or more days of moist curing. In fact, the use of fly ash improved the 50% RCA mixture so much that it achieved one of the highest strength values at later ages. The use of fly ash also improved the strength of the 100% RCA mixture to values similar to those obtained by the fly ash mixture without RCA. Thus, the use of Class C fly ash improved the long-term compressive strength of concretes with RCA content higher than 30%. Despite the differences in strengths, all concrete mixtures developed strengths adequate for concrete pavement construction.

Sudhir P. Patil et. al.[6] (2013) concluded that Natural coarse aggregate used is microtonalite with maximum size of 25 mm. Natural fine aggregate used is river sand and RCA used from demolished concrete waste from which physical test such as specific gravity, absorption and sieve analysis are carried out. Once the optimum mix is determined, it is used to produce concrete with 0%, 25%, 50%, 75% and 100% replacement of RCA. The concrete is prepared to find out the compressive strength and the tensile splitting strength.

The compressive strength of concrete containing50% RCA has strength in close proximity to that of normal concrete. Tensile splitting test shows that concrete has good tensile strength when replace up to 25-50%.Water absorption of RCA is higher than natural aggregate.

Yasser Abdelghany Fawzy[11] (2016) investigated the effect of recycled gravel obtained from low or medium concrete grade on fresh property of concrete (slump), mechanical properties (compressive splitting tensile strength) and mass transport properties (ISAT - sorptivity) of concrete containing dolomite as a natural coarse aggregate. Concrete specimens were prepared with cement, water, sand and dolomite admixed with recycled gravel. The percentage of recycled gravel/dolomite was 0:100, 25:75, 50:50 and 75:25 at w/c =0.50, 0.55 and 0.60.

The results indicated that, increasing the percentage of recycled gravel/dolomite led to decreasing the slump. All mechanical properties of concrete discussed were inversely affected by increasing percentage of recycled gravel/ dolomite from low and medium concrete. Adding 10% SF or bonding admixture increased the mechanical properties of concrete.

IvetaNováková et. al.[12] (2016) concluded that Recycled concrete aggregates (RCA) can be used as a replacement of natural aggregates for a concrete production to save natural sources and also to decrease amount of demolition waste which has to be landfilled. Precast production generates some percentage of defected elements which are carted off and recycled with other demolition waste. In this study defected elements are separately recycled into RCA with beneficial properties and used directly into new mixtures for precast elements.

Results from testing of RCA and application into new concrete mixtures are presented. It was proved that the replacement of raw aggregates by RCA up to 20% has no negative influence on physico-mechanical properties of concrete.

Akash Rao et. al.[13] discussed different aspects of the problem beginning with a brief review of the international scenario in terms of C&D waste generated, recycled

aggregates (RA) produced from C&D waste and their utilization in concrete and governmental initiatives towards recycling of C&D waste. Construction and Demolition (C&D) waste constitutes a major portion of total solid waste production in the world, and most of it is used in land fills. Research by concrete engineers has clearly suggested the possibility of appropriately treating and reusing such waste as aggregate in new concrete, especially in lower level applications.

Use of recycled aggregates in concrete provides a promising solution to the problem of C&D waste management. Based on a survey of production and utilization of RA in RAC, and the properties of RA and RAC discussed in this paper, it is clear that RAC can be used in lower end applications of concrete.

Brett Tempest et. al.[14] investigate the use of crushed construction and demolition waste as a recycled aggregate in the production of new concrete has been successfully demonstrated by researchers as well as by practitioners in the field.

In expanding urban areas, the intensive construction of new infrastructure, as well as rehabilitation and retrofitting of existing infrastructure, opens many potential markets for RAC produced in various grades, including non-structural hard scaping, pavements and even structural applications. The goal of this study was to show that use of recycled aggregates in concrete is

Recycled aggregates produced from construction and demolition rubble can successfully be used in concrete mixtures that exhibit acceptable laboratory performance, impediments to widespread usage are still readily apparent from a market perspective. Due to the ample supply of virgin aggregates in North Carolina, the stream of available recycled aggregates is overwhelmingly directed to lower grade, non-concrete applications. Concrete that includes recycled aggregates has been shown to provide cost savings to producers. If the supply and consistency of demolition rubble increases, there should be improved market interest in RA.

M. Etxeberria et. al.[15] focuses on the possibility of the use of recycled aggregate concrete as a structural material. For that purpose an experimental study of the shear

behaviour and strength of beams made with recycled aggregate concrete was studied. Twelve beam specimens with the same compression strength, four concrete mixtures using different percentages of recycled coarse aggregates (0%, 25%, 50% and 100%) and three different transverse reinforcement arrangements were cast and tested up to failure.

Recycled Aggregate Concrete as Structural Material. The use of recycled aggregates in concrete opens a whole new range of possibilities in the reuse of materials in the building industry. The utilization of recycled aggregates is a good solution to the problem of an excess of waste material, provided that the desired final product quality is reached. The studies on the use of recycled aggregates have been going on for 50years. In fact, none of the results showed that recycled aggregates are unsuitable for structural use. However, some hypothetical problems related to durability aspects resulted in recycled aggregates being employed practically only as base filler for road construction.

The results obtained indicate that a substitution of less than 25% of coarse aggregate, scarcely affects the shear capacity of RC beams, provided that all measures related to dosage and durability aspects have been adopted.

S. K. Singh et. al.[16] presented the experimental results of recycled coarse aggregate concrete and results are compared with the natural crushed aggregate concrete. The fine aggregate used in the concrete, i.e. recycled and conventional is 100 percent natural. The recycled aggregate are collected from four sources all demolished structures. For both types of concrete i.e. M-20 and M-25, w/c ratio, maximum size of aggregate and mix proportion are kept constant.

The development of compressive strength of recycled aggregate concrete at the age of 1, 3, 7, 14, 28, 56, and 90 days; the development of tensile & flexural strength at the age of 1,3,7,14 and static modulus of elasticity at the age of 28 days are investigated. The results shows the compressive, tensile and flexural strengths of recycled aggregate are on average 85% to 95% of the natural aggregate concrete. The durability parameters are also investigated for recycled aggregate concrete and are found to be in good agreement with BIS specifications.

Recycling and reuse of building wastes have been found to be an appropriate solution to the problems of dumping hundreds of thousands tons of debris accompanied with shortage of natural aggregates. Recycled aggregate possess relatively lower bulk density, crushing and impact values and higher water absorption as compared to natural aggregate. The compressive strength of recycled aggregate concrete is relatively lower up to 15% than natural aggregate concrete. The variation also depends on the original concrete from which the aggregates have been obtained. The subject of use of RCA in construction works in India should be given impetus, because of big infrastructural projects are being commissioned including Common Wealth Games in 2010.

Nik. D. Oikonomou[17] refers to the concrete recycling subject and, more specifically, to a proposal for Greek specifications of recycled concrete aggregates (RCA) with reference to international experience and practice. In the present work a guidance of tests and limits of RCA is proposed in order to be used as a basis for pilot and long scale works where the use of RCA can be estimated as more economic and friendlier to the environment. In practical terms, for Greece an experimental use of RCA for a pilot structure is proposed during a first phase, with the addition of only aggregate up to 30% which should comply with the proposed specifications.

2.3 RESEARCH GAP

The work on Replacement of aggregate with percentage variation has been done. But for various percentage combinations it is not studied with varying the age of the structure for various percentage replacements. So, the objectives of dissertation are proposed to perform following objectives in the dissertation to overcome the gaps in the past and present study.

1 To develop the comparison between the strengths of concrete of different parent materials
2 To develop the relationship between compressive strength and flexural strength of concrete.

CHAPTER 3

PROBLEM STATEMENT AND METHODOLOGY

This chapter explains the significance of choosing the problem for experimental investigation and methodology adopted for the same.

3.1 Significance of Present Study

It was also observed that there was lack of uniformity between the types, source and age of the recycled aggregate which were used for the experimental work by the previous researchers. In the proposed work, the building age from which the demolished material is obtained to be used as a coarse aggregate, is chosen to be the parameter of interest. RCA of different sites are chosen and used in different proportions with respect to the natural aggregates. The strength enhancement or diminutions observed against the same. In addition, feasible proportion of RCA is obtained, and the economic point of using RCA is explained.

3.2 Methodology Adopted

3.2.1 Material Collection

Recycled aggregates are to be collected from different sites such that it should be categorized in different years of its life span. This is because as time passes concrete is deteriorated due to various factors such as acid attacks.

B1 – 7 years,

Address – Near Flying Officer SudhirPawar Ln, NarveerTanajiWadi, Shivajinagar, Pune, Maharashtra 411005.

Site available was a building built in 2009. It was previously planned for a 3Star Hotel, but later changed the purpose to a car showroom.

B2 – 21 years,

Address – Sector 18A, New Panvel East, Panvel,
Navi Mumbai, Maharashtra 410206.

Site available was a building built in 1993. The construction work was stopped by the governing body, due to the authorization problem.

B3 – 61 years,

Collected the material from a Demolition contractor.

Address – Near Fergusson College Campus,
Shivajinagar, Pune, Maharashtra 411004

The material supplied was of a building built in 1956, situated in Deccan.

B4 – 12 years,

Address - Near Marathwada Mitra Mandal Vidya Mandi High School,
Kalewadi Main Road, Adarsh Nagar, Kalewadi, Pimpri, Pune, Maharashtra 411017

Site available was a building built in 2005. It was demolished because a new Ring Road was going to be constructed.

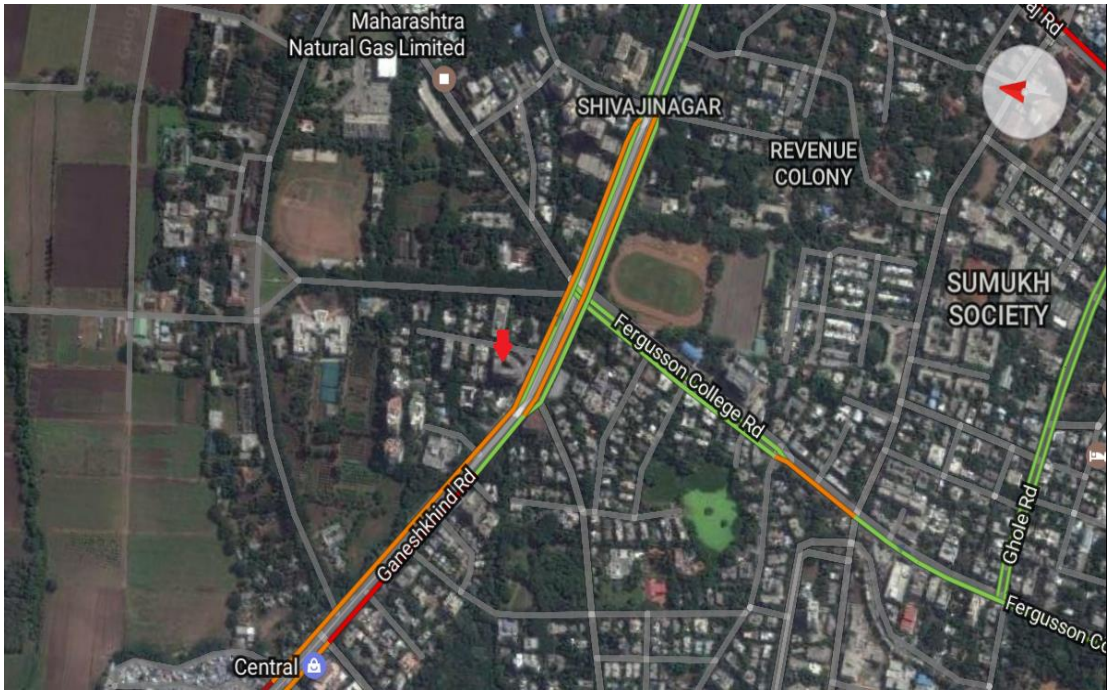


Figure No. 3.1 Site location of B1



Figure No. 3.2 B1 Site



Figure No. 3.3 Site location of B2



Figure No. 3.4 B2 Site

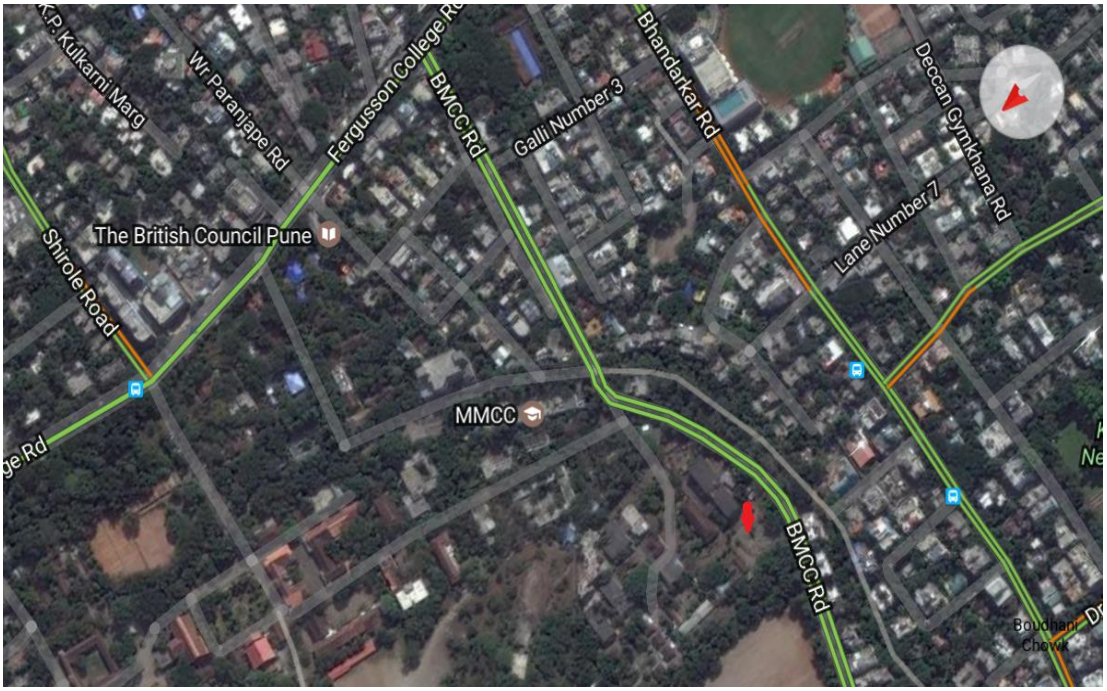


Figure No. 3.5 B3 Site

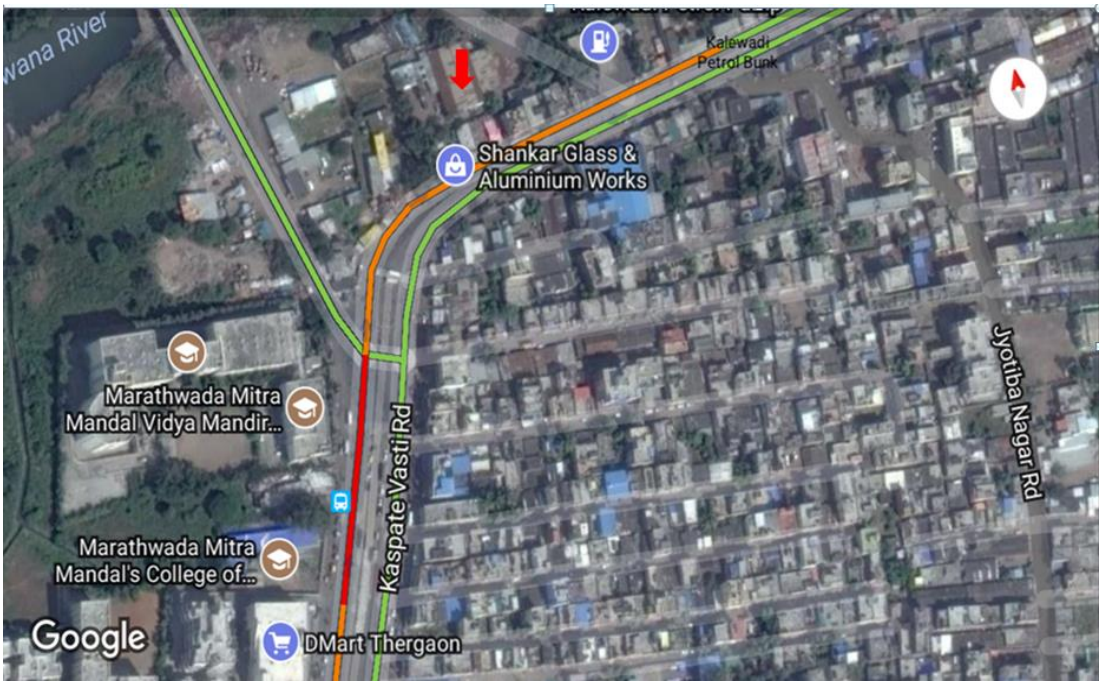


Figure No. 3.6 Site location of B4



Figure No. 3.7 B4 Site

3.2.2 Comparing Basic Properties of Concretes

Concrete of M30 grade was produced, from the material collected from the sites, as it is the majorly used grade.

1. 3 cubes (150 x 150 x 150 mm) of each percentage variation will be casted for Compression Test.
2. 3 Beams (100 x 100 x 500 mm) of each percentage variation will be casted for Flexure Test.

The following properties of concrete were selected for testing:

1. Workability (slump test) immediately after mixing and 30 minutes after mixing,
2. Water absorption (at age of 28 days),
3. Compressive strength f_c (at age of 7 and 28 days),
4. Flexural strength (at age of 28 days),

3.2.3 Safe Percentage Replacement

The natural aggregates are going to be replaced by percentage variation i.e. 10, 20, 30, 40, 50, 60, 70, 80, 90 and 100%. Graph will be plotted with percentage variation on **abscissa** and grade of concrete on **ordinate.** The results of controlled and recycled aggregate are compared.

CHAPTER 4

CONCRETE MATERIAL

4.1 Introduction

The final properties of a material are invariably dependent on the properties of its constituents. The type of constituents can affect the performance of final product to very much extend; hence the material used for the experimental programs are discussed in this chapter.

4.2 Material

1. Cement
2. Water
3. Fine aggregate
4. Coarse aggregate
5. Recycled coarse aggregate

It should be noted that the aggregates in the concrete mix constitute by far the bulk of the mass. The properties of the concrete produced depend upon the amount and type of materials used, and the way they are mixed, handled, compacted, finished and cured.

4.2.1 Properties

1. Cement

There are many types of cement available for making concrete. Some countries have a national standard, which prescribes the requirement for any cement manufactured or used in that country. e.g. Australian Standard AS3972-1991 “Portland and Blended Cements”. Each type of cement produced has a different chemical composition and fineness and gives different properties to a concrete.

There are four main types:-

1. General purpose cement - the most common and the one used for the majority of structures,
2. Type HE - high early strengths cement - gains strength quickly,
3. Type LH - low Heat Cement - produces less heat than GP and Type HE cement. It gains strength more slowly,
4. Type SR - sulphate resisting cement - cement that resists sulphate attack.

2. Aggregates

Coarse aggregates are stones that are more than 5 mm in diameter and are either crushed rock generally from quarries, or gravels excavated from pits or dredged from river beds. Fine aggregates consist of fine and coarse sands or crushed rock finer than 5 mm. The aggregates used should cover a range of sizes. They should be clean and free of any contaminating substances, which may adversely affect the setting time, strength or durability of the concrete, or corrode the steel reinforcement.

Aggregates should not contain:-

1. Weak substances such as pieces of wood, humus or coal
2. Dirt, clay dust or silt coatings.
   (These reduce the bond between aggregates and the cement paste)
3. Water soluble salts such as sulphates or chlorides.

Aggregates should be strong, hard and durable in order to develop the full strength of the cement paste and maximum wear resistance of the concrete. Crumbly or flaky rocks like sandstone, slates or shales should not be used since they lack strength. Chert aggregates should not be used to avoid alkali-silica reaction.

The shape of the aggregate particles is also important since the shape affects both the workability and strength of the concrete. Smooth rounded

aggregates produce workable and easy to handle concrete. Angular materials tend to give a stronger concrete but reduce workability. Flaky and elongated materials promote segregation and reduce workability. They also require more sand and cement. The use of these aggregates should be limited.

There are certain types of aggregates that can react with cement in the presence of moisture. Such reactions result in expansive compounds, which crack and deteriorate the concrete. Suspect aggregates should not be used.

3. Mixing water

Water, which is necessary for hydration, must be clean and fresh and not contain any impurities since these may affect the concrete properties. It is generally accepted that water, which is fit for drinking, is suitable for making concrete. Seawater should not be used in making concrete, particularly reinforced concrete, as it will corrode the steel reinforcement. Bore water must be analyzed first. Industrial, waste or brackish water should not be used. Sugars and detergents have a retarding effect on the setting properties of concrete and, therefore, should not be added to a concrete mix.

4.2.2 Material Testing

4.2.2.1 Cement

Ordinary Portland cement of 43 Grade is used in this project work. The Specific Gravity is 3.0

1. Fineness Test

Weight of cement = 100 gm.
Weight retained on IS 90µ sieve = 7 gm. (After 15 min. of hand shaking)
Percentage residue by weight = 7 %

As per IS 4031 – 1968, the residue should not exceed 7 % by weight of the cement ample. The test is below the limiting value.

2. Standard Consistency Test

The Standard Consistency of the cement is 35.3 %

3. Setting Time

Initial setting time of cement = 42 min. (> 30 min.)

Final setting time of cement = 520 min. (< 600 min.)

4. Compressive Strength test of Cement–Mortar Cube



Figure No. 4.1 Cement Grade Test

Table No. 4.1 Cement–Mortar Cube Compressive Strength

Curing Period	Average Compressive Strength (N/mm^2)	Value according to IS 1226 – 2009 (N/mm^2)
28 days	52.5	> 43

Results of compressive strength of mortar cubes are as per IS specifications. So this cement is OK to use in concrete for experimental study.

5. Soundness test

Difference between the two pointers should not exceed 10 mm as per IS. The soundness of the cement sample is found to be 1 mm. the test results fall within the limit.

4.2.2.2 Coarse Aggregate

Various tests on coarse aggregate were done as per IS 383 – 1970.

1. Specific Gravity of coarse aggregate

The minimum size of coarse aggregate taken is 10 mm.

Weight of aggregate with bucket immersed in water (W_1) = 3359 gm.

Weight of bucket immersed in water (W_2) = 2741 gm.

Weight of saturated surface dried aggregate (W_3) = 997gm.
(After this aggregates were place in oven for drying for 24 hours)

Weight of oven dried aggregate (W_4) = 982gm.

$$\text{Specific gravity (S)} = \frac{W_4}{W_3 - (W_1 - W_2)} \quad \ldots (4.1)$$

$$= \frac{982}{997 - (3359 - 2741)} = \mathbf{2.59\ gm/cc.}$$

2. Water absorption

$$\text{Water absorption} = \frac{W_3 - W_4}{W_4} \times 100 \quad \ldots (4.2)$$

$$= \frac{997 - 982}{982} \times 100 = \mathbf{1.52\ \%}$$

4.2.2.3 Fine Aggregates

The grading must be uniform throughout the work. The moisture content or absorption characteristics must be closely monitoring. Size of fine aggregate should be bigger than 125µ and smaller than 4.75 mm.

1. Specific Gravity of fine aggregate

Weight of Pycnometer, sand and water (W_1) = 1815 gm.

Weight of Pycnometer and water (W_2)= 1523 gm.

Weight of saturated surface dried aggregate (W_3) = 507 gm.
(After this aggregates were place in oven for drying for 24 hours)

Weight of oven dried aggregate (W_4) = 499 gm.

$$\text{Specific gravity (S)} = \frac{W_4}{W_3 - (W_1 - W_2)} \quad \ldots (4.3)$$

$$= \frac{499}{507 - (1815 - 1523)} = \mathbf{2.32 gm/cc.}$$

2. Water absorption

$$\text{Water absorption} = \frac{W_3 - W_4}{W_4} \times 100$$

$$= \frac{507 - 499}{499} \times 100 = \mathbf{1.60\ \%}$$

4.2.2.4 Water

Clean potable water was used for mixing.

CHAPTER 5

INVESTIGATIONS

5.1 Introduction

The details of experimental program are elaborated in this chapter. Casting of specimens i.e. cubes and beams and the determination of its strengths is covered in this chapter.

The concrete mix design is been shown in ANNEXURE A. It is been summarized below:

Table No. 5.1 Mix proportion for M 30 Grade of Concrete

Description	Cement	Fine Aggregate	Coarse Aggregate	Water
Mix Proportion (by weight)	1	0.74	2.92	w/c = 0.38
Quantities of Material (kg/m^3)	466	346.11	1362.01	177

Table No. 5.2 Mix proportion and Concrete batches

Percentage Variation	Cement (kg/m^3)	Aggregate (kg/m^3)			Water (lit/m^3)
		Fine	Coarse	Recycled	
10	466	346.11	1225.81	136.20	177
20	466	346.11	1089.61	272.40	177
30	466	346.11	953.41	408.60	177
40	466	346.11	817.21	544.80	177
50	466	346.11	681.00	681.00	177
60	466	346.11	544.80	817.21	177
70	466	346.11	408.60	953.41	177
80	466	346.11	272.40	1089.61	177
90	466	346.11	136.20	1225.81	177
100	466	346.11	0	1362.01	177

As can be seen above, only natural coarse aggregate was replaced with recycled coarse aggregate. Else was kept the same.

5.2 Specimen Preparation

5.2.1 Cube

An experimental program was conducted to investigate the response behaviour of concrete cube with recycled aggregate. The test specimen can be divided into three categories as per life span of the parent concrete.

Further each category was sub-divided as per percentage variation given to the coarse aggregate. The total 246 cubes were casted of M30 grade as per IS 10262 – 1982, i.e. 60 cubes of B1, 60 cubes of B2, 60 cubes of B3, 60 cubes of B4 and 6 cubes of conventional concrete. The specimen consists of size 150 mm x 150 mm x 150 mm.

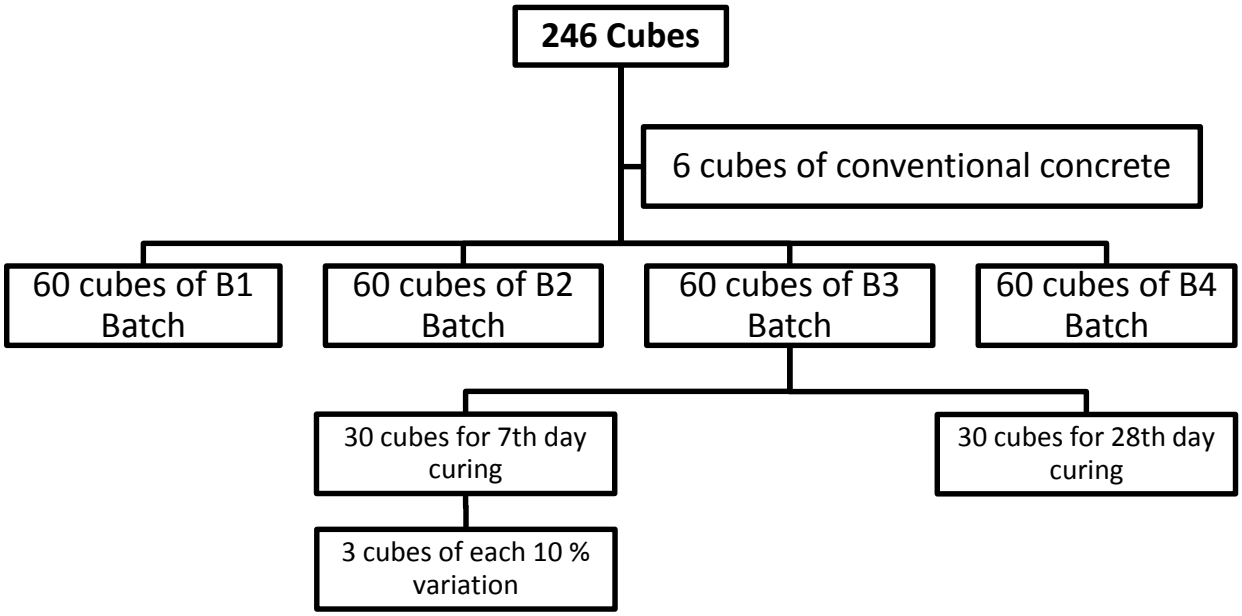


Figure No. 5.1 Flow chart of specimen

5.2.2 Beam

An experimental program was conducted to investigate the response behaviour of concrete beam with recycled aggregate. The test specimens were divided similarly

to that of cubes. The total 246 beam were casted of M30 grade as per IS 10262 – 1982. The specimen consists of size 100 mm x 100 mm x 500 mm.



Figure No. 5.2 Mixing of Concrete



Figure No. 5.3 Casting of cube



Figure No. 5.4 Casting of beam



Figure No. 5.5 Specimens Casted

5.3 Curing

Curing was done to prevent the loss of water which is essential for the process of hydration and hence for hardening. It also prevents the exposure of concrete to a hot atmosphere which might have led to quick drying out of moisture in the concrete and there by subjected it to contraction stresses at a stage when the concrete would not be strong enough to resist them. Here curing was done by keeping the specimen submerged

in the water tank for the period of 7 days and 28 days. After the curing period was over the specimen were taken out of the water tank and dried for a day.



Figure No. 5.6 Curing Tank

5.4 Experimental Setup for Cube

The accuracy of the testing machine shall be such that the percentage of error for loads within the proposed range of use of the testing machine shall not exceed ± 1.0 percent of the indicated load. The indicated load of a testing machine shall not be corrected either by calculation or by the use of calibration diagram to obtain values within the required permissible variation.

The measured compressive strength of the specimen shall be calculated by dividing the maximum load applied to the specimen during the test by the cross-sectional area. Calculated for the entire mean dimensions of the section (IS 1199 – 1959) and shall be expressed to the nearest kg/sq.cm. Average of the three values shall be taken as the

representative of the batch provided the individual variation is not more than ± 15 percent of the average. Otherwise repeat tests shall be made.

The cubes were tested on Compression testing machine (CTM) of 250 ton capacity. The testing procedure adopted for all specimens was same.



Figure No. 5.7 Drying of Cube



Figure No. 5.8(a) Testing of Cube



Figure No. 5.8(b) Testing of Cube

5.5 Experimental Setup for Beam

The cubes were tested on Universal testing machine (UTM) of 60 ton capacity. The testing procedure adopted for all specimens was same.

The measuring strength of specimen is calculated by the following formula,

$$\frac{3LF}{2bd^2} \quad \ldots (5.1)$$

Where,

L - Length of specimen

F - Total force applied to the specimen

b - Width of specimen

d - Thickness of specimen



Figure No. 5.9 Testing of Beam



Figure No. 5.10 Specimen after testing

CHAPTER 6

RESULT AND DISCUSSION

In this experimental program, the behaviour of the concrete with recycled aggregate. The cubes were tested till ultimate load failure and beam were tested till failure in flexural bending. In this chapter the experimental results of all the cubes and beams are interpreted. Their behaviour throughout the test is described using data on flexural bending and the ultimate load carrying capacity.

6.1 Results of Cube Experimental Program

The results of compressive strength of conventional concrete after 7 days and 28 days are presented in Table 7.1 and Table 7.2 respectively.

Table No. 6.1 Compressive Strength for Conventional Concrete at 7th Day

Cube No.	Compressive Strength	Average Compressive Strength
C1	21.24N/mm^2	21.76 N/mm^2
C2	22.31N/mm^2	
C3	21.73N/mm^2	

Table No.6.2 Compressive Strength for Conventional Concrete at 28th Day

Cube No.	Compressive Strength	Average Compressive Strength
C4	32.39N/mm^2	21.76 N/mm^2
C5	32.57N/mm^2	
C6	32.33N/mm^2	

Table No. 6.3 Compressive Strength of B1 at 7th Day

Cube No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
CR1	10	18.59	18.64
		18.28	
		19.04	
CR2	20	19.79	19.51
		19.62	
		19.13	
CR3	30	18.22	18.12
		18.03	
		18.10	
CR4	40	17.96	17.90
		17.43	
		18.32	
CR5	50	15.62	16.02
		16.28	
		16.17	
CR6	60	13.64	13.45
		12.99	
		13.72	
CR7	70	11.41	11.93
		12.36	
		12.01	
CR8	80	11.92	11.54
		10.86	
		11.83	
CR9	90	9.48	9.63
		9.36	
		10.06	
CR10	100	6.34	7.21
		8.26	
		7.03	

Table No.6.4 Compressive Strength of B1 at 28th Day

Cube No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
CR11	10	29.06	28.24
		27.13	
		28.52	
CR12	20	24.88	24.29
		23.93	
		24.06	
CR13	30	25.21	24.68
		24.26	
		24.58	
CR14	40	25.54	25.06
		24.93	
		24.71	
CR15	50	23.09	23.19
		22.81	
		23.66	
CR16	60	19.84	19.72
		19.53	
		19.79	
CR17	70	19.39	19.21
		19.14	
		19.10	
CR18	80	18.69	17.95
		16.77	
		18.40	
CR19	90	15.39	15.26
		16.06	
		14.32	
CR20	100	13.93	13.47
		13.67	
		12.81	

Table No. 6.5 Compressive Strength of B2 at 7th Day

Cube No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
CR21	10	18.98	20.45
		21.01	
		21.36	
CR22	20	20.38	21.14
		22.09	
		20.95	
CR23	30	19.96	19.87
		20.03	
		19.61	
CR24	40	19.91	19.36
		19.22	
		18.94	
CR25	50	18.91	18.97
		19.41	
		18.59	
CR26	60	17.11	17.23
		17.01	
		17.58	
CR27	70	16.31	16.04
		15.84	
		15.98	
CR28	80	13.41	13.27
		13.52	
		12.88	
CR29	90	13.51	12.18
		11.52	
		11.51	
CR30	100	10.77	10.01
		11.07	
		8.19	

Table No.6.6 Compressive Strength of B2 at 28th Day

Cube No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
CR31	10	28.99	29.58
		30.14	
		29.62	
CR32	20	32.16	31.55
		31.62	
		30.87	
CR33	30	29.97	30.48
		30.36	
		31.10	
CR34	40	28.10	28.41
		28.64	
		28.49	
CR35	50	26.11	26.19
		26.38	
		26.07	
CR36	60	26.33	25.93
		25.24	
		26.22	
CR37	70	21.27	21.72
		22.57	
		21.33	
CR38	80	16.36	16.46
		17.59	
		15.44	
CR39	90	16.06	16.22
		16.26	
		16.34	
CR40	100	15.18	14.27
		12.96	
		14.68	

Table No. 6.7 Compressive Strength of B3 at 7th Day

Cube No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
CR41	10	16.59	16.58
		17.78	
		15.69	
CR42	20	16.07	15.88
		15.44	
		15.82	
CR43	30	10.00	10.74
		11.14	
		11.04	
CR44	40	6.86	7.21
		7.29	
		7.46	
CR45	50	7.29	6.99
		6.86	
		6.86	
CR46	60	6.29	6.41
		6.45	
		6.61	
CR47	70	5.89	5.72
		5.82	
		5.59	
CR48	80	5.07	5.14
		5.36	
		4.93	
CR49	90	5.07	4.87
		4.79	
		4.65	
CR50	100	3.99	4.16
		4.38	
		4.25	

Table No. 6.8 Compressive Strength of B3 at 28th Day

Cube No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
CR51	10	25.15	24.55
		24.30	
		24.20	
CR52	20	18.46	19.82
		20.29	
		20.72	
CR53	30	17.76	17.64
		17.28	
		17.88	
CR54	40	15.69	15.93
		15.76	
		16.33	
CR55	50	11.79	11.72
		11.25	
		12.12	
CR56	60	9.63	10.09
		9.50	
		11.14	
CR57	70	7.55	7.99
		8.08	
		8.33	
CR58	80	7.72	7.57
		7.44	
		7.55	
CR59	90	7.29	7.17
		7.28	
		6.95	
CR60	100	6.57	6.12
		5.97	
		5.82	

Table No. 6.9 Compressive Strength of B4 at 7th Day

Cube No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
CR51	10	21.77	22.04
		22.41	
		21.94	
CR52	20	20.64	20.45
		20.13	
		20.57	
CR53	30	19.68	19.86
		19.76	
		20.13	
CR54	40	19.56	19.58
		19.43	
		19.74	
CR55	50	20.11	20.17
		20.13	
		20.26	
CR56	60	18.71	18.67
		18.49	
		18.82	
CR57	70	17.69	17.54
		17.34	
		17.58	
CR58	80	16.47	16.80
		16.72	
		17.22	
CR59	90	15.55	15.78
		15.94	
		15.86	
CR60	100	16.18	16.08
		15.89	
		16.16	

Table No. 6.10 Compressive Strength of B4 at 28th Day

Cube No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
CR51	10	30.46	30.55
		30.61	
		30.59	
CR52	20	29.82	29.61
		29.53	
		29.48	
CR53	30	28.41	28.57
		28.67	
		28.63	
CR54	40	31.59	31.69
		31.64	
		31.83	
CR55	50	29.04	29.16
		29.36	
		29.07	
CR56	60	28.39	28.31
		28.24	
		28.29	
CR57	70	27.72	27.73
		27.80	
		27.66	
CR58	80	25.16	25.18
		25.23	
		25.14	
CR59	90	24.89	24.92
		25.06	
		24.81	
CR60	100	23.44	23.51
		23.61	
		23.48	

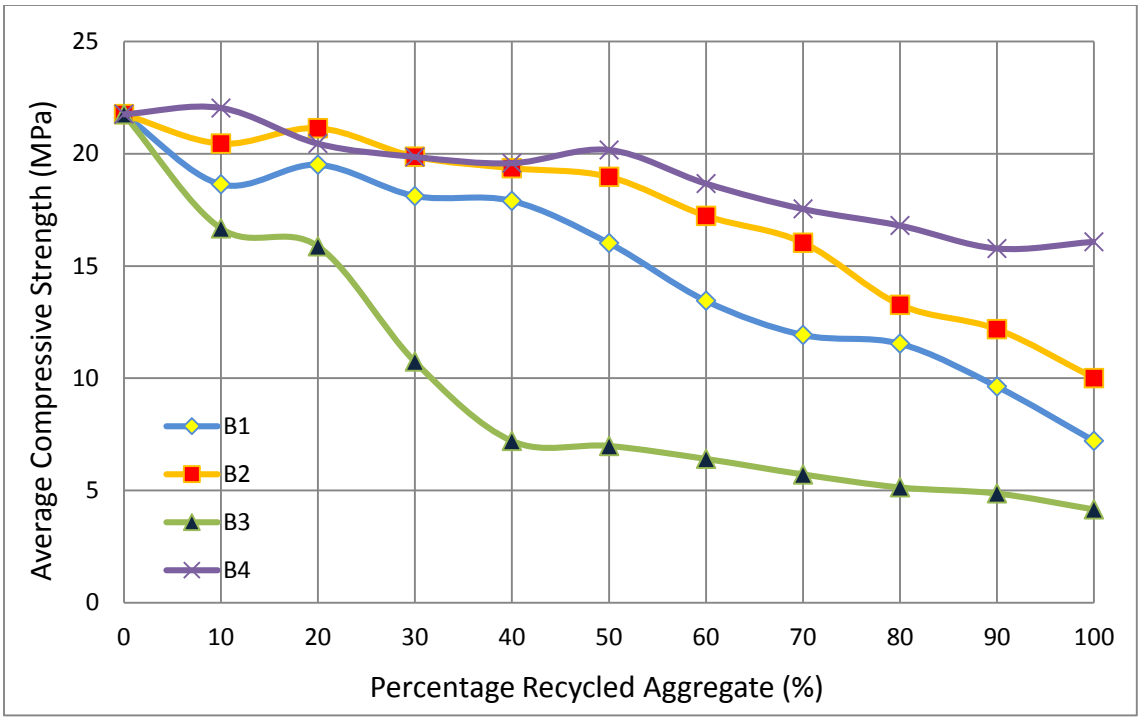


Figure No. 6.1 Comparison of Compressive Strengths of B1, B2, B3 and B4 at 7th day

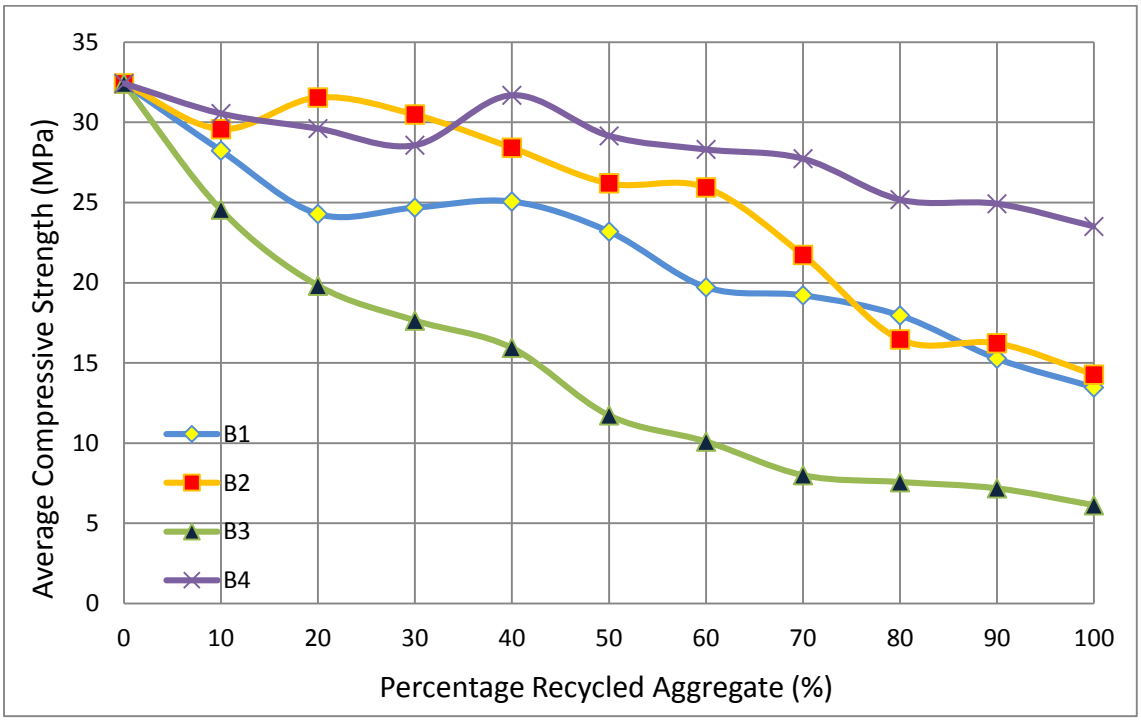


Figure No. 6.2 Comparison of Compressive Strengths of B1, B2, B3 and B4 at 28th day

From the results obtained, it was observed that the results of compressive strength of recycled aggregate concrete is satisfactory during early stage of replacement but it exhibits lower compressive strength during later stage of replacement.

The Figure 6.1 compares the compressive strengths on 7th day of the concretes made of B1 aggregates, B2 aggregates, B3 aggregate and B4 aggregate. It was seen that the concrete gives a satisfactory value of compressive strength till 20% replacement of natural aggregate by recycled aggregate.

Similarly, Figure 6.2 compares the compressive strengths on 28th day of the concretes made of B1 aggregates, B2 aggregates, B3 aggregate and B4 aggregate. Here it came across that results of compressive strength of concrete up to 30% replaced aggregate were satisfactory.

It was found that aggregate obtained from B1 site, have not just solid aggregate but a large proportion of harden mortar. As harden mortar cannot bear load, the complete concrete collapses at lower compressive strength. But as we use older concrete, the aggregate obtained from it have lesser proportion of harden mortar. Deterioration of concrete is more as the time passes. This gave much better results than B1.

This can be corrected by using proper cleaning techniques, crushing techniques and casting methods under highly skilled supervision.

6.2 Results of Beam Experimental Program

The results of flexural strength of conventional concrete after 7 days and 28 days are presented in Table 7.9 and Table 7.10 respectively.

Table No. 6.11 Flexural Strength for Conventional Concrete at 7th Day

Beam No.	Flexural Strength	Average Flexural Strength
B1	2.88 N/mm^2	2.90 N/mm^2
B2	2.91 N/mm^2	
B3	2.90 N/mm^2	

Table No. 6.12 Flexural Strength for Conventional Concrete at 28th Day

Beam No.	Flexural Strength	Average Flexural Strength
B4	3.58 N/mm^2	3.57 N/mm^2
B5	3.54 N/mm^2	
B6	3.60 N/mm^2	

Table No. 6.13 Flexural Strength of B1 at 7th Day

Beam No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
BR1	10	2.88	2.90
		2.91	
		2.90	
BR2	20	2.71	2.68
		2.69	
		2.64	
BR3	30	2.74	2.75
		2.77	
		2.73	
BR4	40	2.66	2.68
		2.68	
		2.69	
BR5	50	2.61	2.63
		2.64	
		2.63	
BR6	60	2.42	2.41
		2.40	
		2.41	
BR7	70	2.26	2.28
		2.29	
		2.30	
BR8	80	2.06	2.14
		2.18	
		2.17	
BR9	90	2.10	2.11
		2.09	
		2.13	
BR10	100	1.89	1.84
		1.83	
		1.81	

Table No. 6.14 Flexural Strength of B1 at 28th Day

Beam No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
BR11	10	3.58	3.57
		3.54	
		3.60	
BR12	20	3.32	3.33
		3.33	
		3.35	
BR13	30	3.08	3.09
		3.09	
		3.09	
BR14	40	3.14	3.11
		3.10	
		3.10	
BR15	50	3.11	3.13
		3.15	
		3.12	
BR16	60	3.00	3.02
		3.03	
		3.04	
BR17	70	2.76	2.78
		2.79	
		2.79	
BR18	80	2.73	2.74
		2.78	
		2.71	
BR19	90	2.65	2.65
		2.66	
		2.64	
BR20	100	2.44	2.45
		2.48	
		2.43	

Table No. 6.15 Flexural Strength of B2 at 7th Day

Beam No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
BR21	10	2.89	2.90
		2.89	
		2.91	
BR22	20	2.81	2.81
		2.83	
		2.80	
BR23	30	2.83	2.87
		2.92	
		2.87	
BR24	40	2.72	2.76
		2.79	
		2.76	
BR25	50	2.72	2.70
		2.71	
		2.68	
BR26	60	2.72	2.71
		2.73	
		2.69	
BR27	70	2.59	2.58
		2.59	
		2.57	
BR28	80	2.50	2.51
		2.52	
		2.52	
BR29	90	2.24	2.26
		2.26	
		2.27	
BR30	100	2.23	2.17
		2.13	
		2.16	

Table No. 6.16 Flexural Strength of B2 at 28th Day

Beam No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
BR31	10	3.56	3.57
		3.57	
		3.56	
BR32	20	3.37	3.41
		3.44	
		3.41	
BR33	30	3.55	3.52
		3.52	
		3.48	
BR34	40	3.43	3.46
		3.45	
		3.49	
BR35	50	3.32	3.34
		3.35	
		3.34	
BR36	60	3.20	3.20
		3.22	
		3.20	
BR37	70	3.21	3.19
		3.15	
		3.21	
BR38	80	2.89	2.92
		2.97	
		2.89	
BR39	90	2.53	2.54
		2.63	
		2.46	
BR40	100	2.51	2.52
		2.52	
		2.53	

Table No. 6.17 Flexural Strength of B3 at 7th Day

Beam No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
BR41	10	2.55	2.56
		2.64	
		2.48	
BR42	20	2.51	2.49
		2.46	
		2.49	
BR43	30	1.98	2.05
		2.09	
		2.08	
BR44	40	1.64	1.68
		1.69	
		1.71	
BR45	50	1.69	1.66
		1.64	
		1.64	
BR46	60	1.57	1.59
		1.59	
		1.61	
BR47	70	1.52	1.50
		1.51	
		1.48	
BR48	80	1.41	1.42
		1.45	
		1.39	
BR49	90	1.41	1.38
		1.37	
		1.35	
BR50	100	1.25	1.28
		1.31	
		1.29	

Table No. 6.18 Flexural Strength of B3 at 28th Day

Beam No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
BR51	10	3.14	3.10
		3.09	
		3.08	
BR52	20	2.69	2.79
		2.82	
		2.85	
BR53	30	2.81	2.63
		2.53	
		2.55	
BR54	40	2.48	2.50
		2.49	
		2.53	
BR55	50	2.15	2.14
		2.10	
		2.18	
BR56	60	1.94	1.99
		1.93	
		2.09	
BR57	70	1.72	1.77
		1.78	
		1.81	
BR58	80	1.74	1.72
		1.69	
		1.72	
BR59	90	1.69	1.68
		1.71	
		1.65	
BR60	100	1.62	1.55
		1.53	
		1.51	

Table No. 6.19 Flexural Strength of B4 at 7th Day

Beam No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
BR51	10	2.92	2.92
		2.96	
		2.88	
BR52	20	2.84	2.81
		2.79	
		2.80	
BR53	30	2.76	2.77
		2.76	
		2.79	
BR54	40	2.74	2.75
		2.76	
		2.76	
BR55	50	2.78	2.79
		2.79	
		2.81	
BR56	60	2.71	2.69
		2.68	
		2.67	
BR57	70	2.59	2.60
		2.60	
		2.60	
BR58	80	2.54	2.55
		2.56	
		2.54	
BR59	90	2.47	2.47
		2.47	
		2.46	
BR60	100	2.52	2.49
		2.45	
		2.51	

Table No. 6.20 Flexural Strength of B4 at 28th Day

Beam No.	Percentage Variation (%)	Compressive strength (N/mm^2)	Average Compressive Strength (N/mm^2)
BR51	10	3.46	3.46
		3.46	
		3.46	
BR52	20	3.42	3.41
		3.40	
		3.40	
BR53	30	3.34	3.35
		3.35	
		3.35	
BR54	40	3.52	3.52
		3.52	
		3.53	
BR55	50	3.37	3.38
		3.39	
		3.38	
BR56	60	3.34	3.33
		3.33	
		3.33	
BR57	70	3.30	3.30
		3.30	
		3.29	
BR58	80	3.14	3.14
		3.14	
		3.14	
BR59	90	3.12	3.13
		3.13	
		3.12	
BR60	100	3.03	3.04
		3.04	
		3.03	

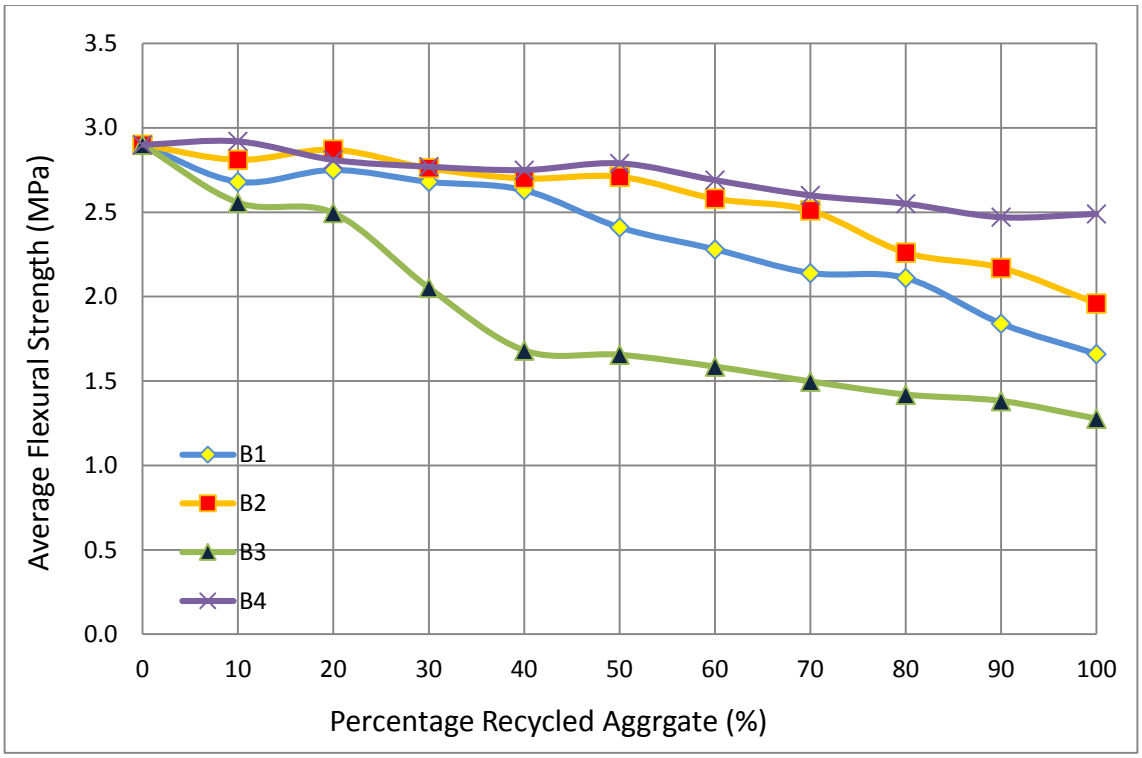


Figure No. 6.3 Comparison of Flexural Strengths of B1, B2, B3 and B4 at 7th day

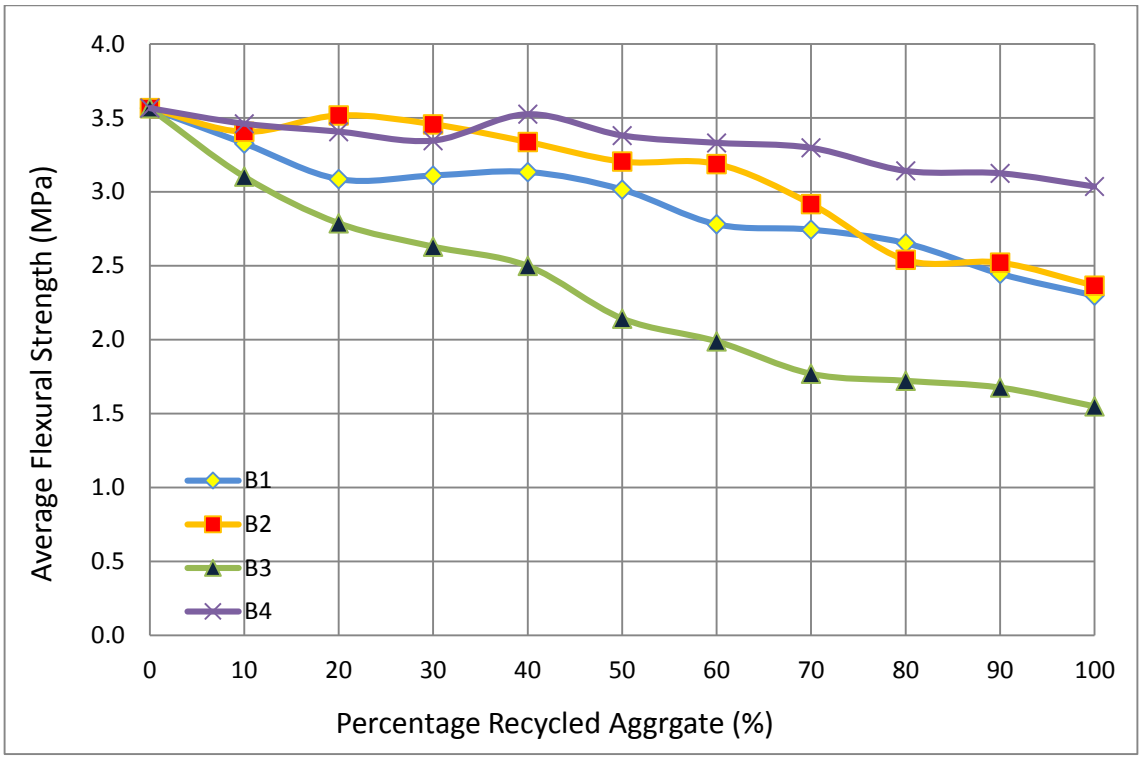


Figure No. 6.4 Comparison of Flexural Strengths of B1, B2, B3 and B4 at 28th day

From the results obtained, it was observed that the results of flexural strength of recycled aggregate concrete is satisfactory during early stage of replacement but it exhibits lower flexural strength during later stage of replacement.

The Figure 6.3 compares the flexural strengths on 7th day of the concretes made of B1 aggregates, B2 aggregates, B3 aggregate and B4 aggregate. It was seen that the concrete gives a satisfactory value of flexural strength till 30% replacement of natural aggregate by recycled aggregate.

Similarly, Figure 6.4 compares the flexural strengths on 28th day of the concretes made of B1 aggregates, B2 aggregates, B3 aggregate and B4 aggregate. Here it came across that results of flexural strength of concrete up to 10% replaced aggregate were satisfactory.

This can be corrected by using proper cleaning techniques and casting methods under highly skilled supervision.

6.3 Study of Cost

To verify the economic viability of the study developed, two comparative budgets have been done.

6.3.1 Coarse Aggregate

Natural Aggregate

Cost of Natural Aggregate = Rs. 2500 per Brass.

Recycled Aggregate

Recycled Aggregate = Transporting CDW to crusher = Rs. 1000 per Brass

Crushing Process = Rs. 500 per Brass

Total = Rs. 1500 per Brass

6.3.2 Fine Aggregate

Natural Aggregate

Cost of Natural Fine Aggregate	=	Rs. 7500 per Brass

Recycled Aggregate

Recycled Fine Aggregate	=	By product of Recycled Coarse Aggregate

Hence it can be seen that using recycled aggregate reduces the cost of construction.

CHAPTER 7

CONCLUSION

7.1 Conclusion

From the experimental work carried out the following conclusion can be drawn:

1. Recycled Aggregate from older building gives better results, as deterioration of concrete is more with time and aggregates obtained are more solid.
2. Workability decreases with the increase in percentage replacement of recycled aggregate.
3. Tests conducted on recycled aggregates concrete and results compared with natural aggregates concrete are satisfactory.
4. Compressive strength and flexural strength of concrete with 30% replaced recycled aggregate were acceptable.
5. It was observed that if we require a particular strength of concrete using recycled aggregate, one should adopt a higher proportion of mix design.
6. Use of recycled aggregate, cost of transportation of natural resources & excavation is considerably saved. This reduces the waste material impact on environment.
7. As the treatment process for recycled aggregate was inadequate, strength achieved were not upto mark, but by applying more advanced treatment process the strength can be improved.

7.2 Future Scope

Following are the future scopes for this project

1. One can investigate by taking different rocks physical as well as chemical properties of the recycled aggregate.
2. The crushed construction and demolition wastes which is of size equal to that of fine aggregate can be used to replace river sand.

3. One can check for different concrete mix design and form a regression equation so as the accuracy can be calculated with respect to the conventional concrete.
4. FRC can be used so as more strength can be achieved

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ANNEXURE I
CONCRETE MIX DIESIGN

IS: 10262 – 2009 furnishes the following steps to arrive at a final mix proportion for a given grade of concrete.

A.I.1 Data required

The data required for the concrete mix proportion is as follow

1.	Grade designation	-	M 30
2.	Type of Cement	-	OPC 43 grade
3.	Maximum size of Aggregate	-	20 mm
4.	Minimum cement content	-	320 kg/m^3
5.	Maximum water - cement ratio	-	0.45
6.	Exposure condition	-	Mild
7.	Type of aggregate	-	Rounded aggregate
8.	Specific gravity of cement	-	3
9.	Specific gravity of F.A.	-	2.32
10.	Specific gravity of C.A.	-	2.59
11.	Fine aggregate	-	Zone III

A.I.2 Target Strength of Mix Proportioning

The target strength of mix proportion is calculated using following formula given in IS: 10262 – 2009, Clause No. 3.2.

$$f_m = f_{ck} + 1.65\sigma \qquad \text{(A.1)}$$

Where, f_m = Target mean compressive strength

f_{ck} = Characteristic compressive strength

σ = Standard deviation

From Table 1 of IS: 10262 – 2009, Standard deviation,

σ = **5 N/mm^2**

Therefore,

Target strength = 30 + (1.65 × 5)

= **38.25 N/mm^2**

A.I.3 Selection of Water - Cement ratio

Table A.I.1 Min. Cement Content, Water-Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregate of 20 mm max. Size Aggregate for Reinforced Concrete

Source: Table 5, IS: 456 - 2000

Exposure	Min. Cement Content (kg/m^3)	Min. Free Water-Cement Ratio	Min. Grade of Concrete
Mild	300	0.55	M 20
Moderate	300	0.50	M 25
Severe	320	0.45	M 30
Very Severe	340	0.45	M 35
Extreme	360	0.40	M 40

Therefore, w/c ratio = 0.45 … (a)

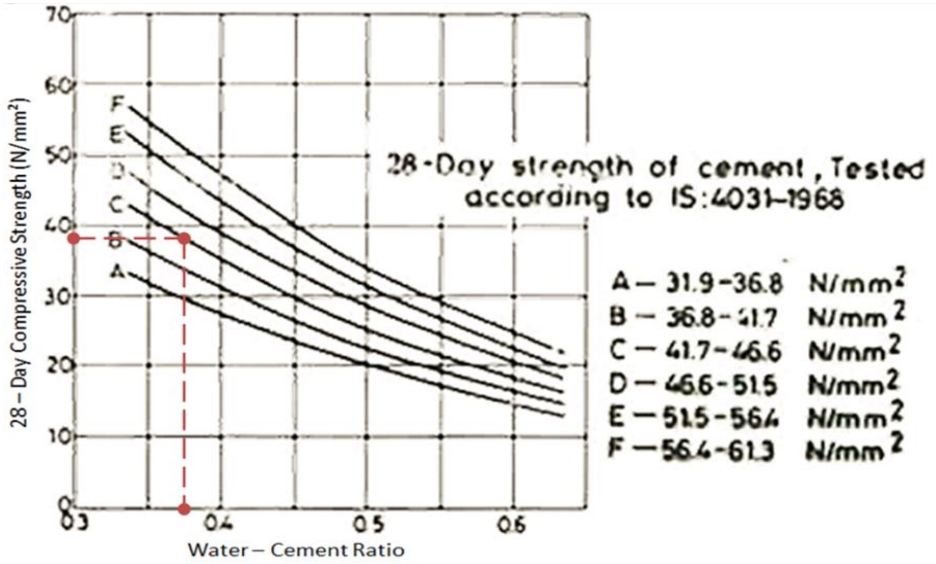


Figure A.I.1Relations between water cement ratio and compressive strength of concrete using crushed aggregates

Source: IS 4031 - 1968

As, Cement strength = 43 N/mm^2, select curve C

Therefore, w/c ratio = 0.38 … (b)

Take minimum of (a) and (b)

Adopted w/c ratio = **0.38**

A.I.4 Selection of Water Content

Table A.I.2 Maximum Water Content per cubic meter of Concrete for Maximum Size of Aggregate

Source: Table 2 and Table 3, IS: 10262 - 2009

Nominal Maximum Size of Aggregate (mm)	Maximum Water Content (kg/m^3)	Total Aggregate (%)
10	208	40
20	186	35
40	165	30

Adopted Water content = **186 kg/m^3** and, Total Aggregate = **35 %**

A.I.5 Adjustments

As the above values taken have the following conditions

1 Crushed Coarse Aggregate

2 Fine Aggregate Zone II

3 Water – Cement Ratio = 0.6

4 Compaction Factor = 0.8

As seen the actual condition different, we need to perform some adjustments.

Table A.I.3 Adjustments provided

Source: Annex A, IS: 10262 - 2009

Condition	Adjustments	
	ΔW	**ΔF_{agg} (%)**
Sand zone I, III and IV	0	-1.5 % per Zone (+ 1.5 for Zone I)
Compaction factor (± 0.1)	± 3 %	0
Water-Cement Ratio (± 0.05)	0	± 1%
Rounded Aggregate	- 15 kg/m^3	- 7 %

Therefore,

$$\Delta W = \left(\frac{0.9 - 0.8}{0.1} \times 0.03 \times 186\right) - 15 = -\mathbf{9.42\ kg/m^3} \quad \text{(A.2)}$$

And, $$\Delta p = -1.5 + \frac{0.38 - 0.6}{0.05} - 7 = -\mathbf{12.9\ \%} \quad \text{(A.3)}$$

Therefore, Final

Water Content = 186 – 9.42 = 176.58 ≈ **177 kg/m^3**

And, Aggregate (C.A. and F.A.) = 35 – 12.9 = **22.1 %**

A.I.6 Calculation of Cement Content

$$\text{Cement Content} = \frac{\text{Water Content}}{\text{Water–Cement ratio}} \quad \text{(A.3)}$$

$$= \frac{177}{0.38} = 465.78 \approx \mathbf{466\ kg/m^3}$$

A.I.7 Calculation of C.A. and F.A. content

For F.A.

$$\mathbf{V} = \frac{\mathbf{1}}{\mathbf{1000}} \times \left\{\mathbf{W} + \frac{\mathbf{C}}{\mathbf{S_C}} + \left(\frac{\mathbf{1}}{\mathbf{p}} \times \frac{\mathbf{F_{agg}}}{\mathbf{S_{F_{agg}}}}\right)\right\} \quad \text{(A.4)}$$

For C.A.

$$\mathbf{V} = \frac{\mathbf{1}}{\mathbf{1000}} \times \left\{ \mathbf{W} + \frac{\mathbf{C}}{\mathbf{S_C}} + \left(\frac{\mathbf{1}}{\mathbf{1-p}} \times \frac{\mathbf{C_{agg}}}{\mathbf{S_{C_{agg}}}} \right) \right\} \quad \text{(A.5)}$$

Where,

V = Volume of concrete = 1 kg/m^3
W = Water Content
C = Cement Content
S_C = Specific Gravity of Cement
p = Total Percentage of Aggregate
F_{agg} = Fine Aggregate Content
C_{agg} = Coarse Aggregate Content
$S_{F_{agg}}$ = Specific Gravity of Fine Aggregate
$S_{C_{agg}}$ = Specific Gravity of Coarse Aggregate

Therefore,

$$1 = \frac{1}{1000} \times \left\{ 177 + \frac{466}{3} + \left(\frac{1}{0.221} \times \frac{F_{agg}}{2.32} \right) \right\}$$

F_{agg} = **346.11 kg/m^3**

$$1 = \frac{1}{1000} \times \left\{ 177 + \frac{466}{3} + \left(\frac{1}{1-0.221} \times \frac{C_{agg}}{2.59} \right) \right\}$$

C_{agg} = **1362.01 kg/m^3**

A.I.8 Final Mix Proportion adopted for M 30 grade of Concrete

The mix proportion thus calculated is shown in table format as below

Table A.I.4 Mix proportion for M 30 Grade of Concrete

Description	Cement	Fine Aggregate	Coarse Aggregate	Water
Mix Proportion (by weight)	1	0.74	2.92	w/c = 0.38
Quantities of Material (kg/m^3)	466	346.11	1362.01	177

ANNEXURE II

TESTING CERTIFICATE

Suman Ramesh Tulsiani Charitable Trust's
SUMAN RAMESH TULSIANI TECHNICAL CAMPUS
FACULTY OF ENGINEERING
Approved by All India Council for Technical Education, Recognized by Govt. of Maharashtra
Affiliated to Savitribai Phule Pune University and Maharashtra State Board of Technical Education
DTE Code : EN6767, SPPU Code : CEGP019380, MSBTE Code : 1729
SUMAN RAMESH TULSIANI CHARITABLE TRUST

Date: 13 June 2017

Ref: SRTTC/2016-17/Civil(Dip)/৪০৭

CERTIFICATION OF TESTING

This is to certify that **Mr. Nikhil Vitthal Mulik,** student of JSPM's RSCOE, Tathawade has carried out his experimental work on the topic "**Influence of Replacement of Recycled Aggregate on Strengths of Concrete**" in the departmental laboratory. The testing period was from January 2017 to June 2017. The testing was performed in the fulfillment of his M.E. Civil (Structural Engineering) Dissertation Work.

This experimental work was performed in engineering and ethical way and the results were obtained in numerical and graphical way.

No. of specimens and details are as follows:

Sr. No.	Specimen Description	No. of Specimen
1.	Cube (150 x 150 x 150 mm)	186
2.	Beam (100 x 100 x 500 mm)	186

SRTTC 1729 KHAMSHET

Hegade S.R 13/7/2017

Yours Faithfully

Head of Department

Head - Diploma
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ANNEXURE III

PUBLICATION

1. The Paper Entitled "**Influence of Replacement of Recycled Aggregate on Strengths of Concrete**" has been presented in Civil Post Graduation Conference 2017 (CIVILPGCON 2017) organized by Flora Institute of Technology, Pune in association with Board of Studies Civil Engineering of Savitribai Phule Pune University, Pune on 13th – 14th June 2017.



Flora Education Society's

Flora Institute of Technology, Pune

Department of Civil Engineering
in association with
Board of Studies in Civil Engineering, Savitribai Phule Pune University

Fourth Post Graduate Conference of Civil Engineering

Certificate

This is to certify that Mr./Ms. MULIK NIKHIL VITTHAL
has presented paper titled Influence of Replacement of Recycled Aggregate on Strengths of Concrete
in CIVILPGCON 2017 held during 13th-14th June 2017 at Flora Institute of Technology, Pune.

Dr. Atul Padalkar	Dr. S. S. Pimplikar	Dr. S. K. Kulkarni	Prof. A. A. Gaikwad
Principal	Conference Chair	Conference Co-Chair	Conference Secretary

2. The Paper Entitled "**Use of Recycled Concrete Aggregate and its Influence on Strength of Concrete"** has been published in International Journal of Trend in Research and Development (IJTRD) in Volume 4 Issue 4, July – August 2017.

